Shubham Subrot Panigrahi, Ph.D., P.Eng.

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Resident Status: **Canadian Citizen (also a U.S.A H1-B visa holder)**

Who am I?

Agri-food Process Automation, Computational and Numerical Modeling, Reaction Engineering and Kinetics, Coupled Reaction–Transport Modeling in Biological Materials, Thermodynamics, Fluid Dynamics, Internet-of-Things (IoT), Machine Learning, High-performance Computing, Imaging and Spectroscopy, Data analytics, Life Cycle Analysis, Techno-economic Analysis, Post-harvest Engineering, Heat and Mass Transfer, Stored-product Protection, Grain Handling and Storage, Biosystems Engineering, Food Process and Safety Engineering, Energy Systems Analysis

Education

April 2021
Doctor of Philosophy,
Adelaide University, Adelaide, Australia
Major: Mechanical and Manufacturing Engineering (Specialization: **Computational Physics-based and Data-driven Modeling in Post-harvest Engineering**)
Supervisors: Dr. Chandra B. Singh and Dr. John M. Fielke

May 2017
Bachelor of Technology Food Process Engineering Department,
National Institute Of Technology, Rourkela, India
Major: Food Process Engineering
Supervisor: Dr. Preetam Sarkar
CGPA: 8.95/10

Research Work Experience

January 2026 - Current

Post-doctoral Research Associate at Department of Food Science, University of Arkansas

Details of work:- **Working at the intersection of circular bioeconomy and food safety engineering using multiphysics and data-driven modeling.**
Project details:
- Apply circular bioeconomy frameworks to various nodes of the food system.
- Conducting sustainability analyses and develop microbial risk assessement machine learning models, including life cycle assessment (LCA) and techno-economic analysis (TEA) of novel and conventional food safety interventions in low (grain and dried fruits)/high (meat products) moisture food systems.
- Collaborating with Department of Biological and Agricultural Engineering faculty on developing efficient grain and specialty crop drying, aeration, and storage management protocols.
- Collaborating with University of Georgia food safety faculty and Center for Scalable and Intelligent Automation in Poultry Processing on developing poultry welfare and quality detection models using multiphysics modeling and data-driven approaches.
- Developing physics-based computational model, automated imaging and robotic system for foreign material/contaminants detection/removal on chicken meat and during grain storage. **Designing a Techno-economic tool kit software for analyzing the automation feasibility in poultry industry.**
- Developing non-chemical and non-thermal treatment systems for de-contamination of low/high moisture food products.
- Maintain operations of the food safety and engineering research laboratory.
- **Co-supervising and Mentoring graduate students (1 senior Undergraduate, 2 Masters and 2 PhD), with emphasis on experimental design, data analysis, modeling, publication, technical expertise, and professional development.**
- Assist in the development of graduate-level courses on food safety engineering and circular bioeconomy.
- Developing competitive extramural proposals and securing research support as Principal Investigator, Key Personnel and collaborator, with awarded projects and funding roles listed below.

Grant details:
- **NSF Translation-to-Practice (TTP-T)**: Key personnel in writing the proposal *'Translating AI-Enabled Thermal Computer Vision into Practice for Foreign Material Detection in Poultry Processing'* (PI: Dr. Dongyi Wang; Co-PI: Dr. Jeyam Subbiah and Dr. Casey Owens; Industry Collaborators: FloVision, Simmons and Tyson Foods) (**USD 1,200,000**) (*Status: Submitted*)

- **Industry Contract**: Led the initiative to successfully develop the University of Arkansas and OPI Systems Inc. contract to design grain isotherm curves (**USD 105,000**) (*Status: Funded*)

- **NC-213 The Andersons Research Grant Program**: Principal Investigator in project titled *'Field Validation of Self-Adaptive In-Bin Drying Coupled with Sparse-Sensor Digital Twin Monitoring of Stored Grain'* (**USD 150,000**) (*Status: Submitted*)

- **NSF Translation-to-Practice (TTP-T)**: Key personnel in writing the proposal *'Translating a Cyber-Physical Digital Twin Model for Autonomous On-Farm Commodity Storage'* (PI: Dr. Kaushik Luthra; Co-PI: Dr. Jeyam Subbiah and Dr. Dirk Maier; Industry Collaborators: OPI Systems Inc., Amber Agriculture Inc.) (**USD 1,199,558**) (*Status: Submitted*)

- **USDA Crop Protection and Pest Management (CPPM)**: Key personnel in writing an integrated (Research and Extension) proposal *'Advancing Ozone as a Systems-Based Organic Fumigation Technology for Bulk Grain Handling'* in collaboration with Oxidation Technologies LLC and USDA Manhattan (PI: Dr. Kaushik Luthra; Co-PI: Dr. Jeyam Subbiah) (**USD 299,948**) (*Status: Not Funded: Medium Priority*)

- **USDA AFRI Strengthening Agricultural Systems (SAS)**: Key personnel in writing an integrated (Research, Education and Extension) proposal *'Multi-Modal Physical Sensing for Quantitative Food Composition Measurement Across Agricultural, Food Processing, and Nutrition Systems'* in collaboration with Kalorie Corporation industry (PI: Dr. Kaushik Luthra; Co-PI: Dr. Jeyam Subbiah) (**USD 4,993,770**) (*Status: Not Funded; Medium Priority*)

October 2024 - December 2025

Post-doctoral Research Assistant at Agriculture and Agri-food Canada, Lethbridge Research and Development Centre (Government of Canada)
Details of work:- **Developed LiDAR, RGB, Multispectral and Hyperspectral imaging application on crop phenotyping using big-data analysis, machine learning and deep learning approaches.**
Project details:
- Developed Multispectral image processing protocol to estimate crop/soil residue cover in cover crops.
- Collaborated with barley, canola, sugarbeet and wheat growers and grain management industries on developing efficient drying, aeration, rehydration and storage management protocols.
- Developed image processing protocol using open-source machine learning algorithms to estimate drybean physiological traits using LiDAR and Multispectral sensing units.
- Developed Raspberry Pi-based hyperspectral camera system to predict pea root rot caused by *Apanomyces* and detect wheat minerals. Used deep learning approaches in image processing and feature extraction.
- Assisted in organizing digital agriculture field day and gave a technical talk on the benefits of using imaging applications for crop production.

May 2021 - September 2024

Post-doctoral Research Associate at Lethbridge Polytechnic, Canada

Details of work:- **Completed four projects directed towards Prairie farmers and industries (OPI Systems and Lantic Sugars), supervised and mentored students and postdocs, and written project and laboratory development proposals**.
Project details:
- Developed on-farm barley, canola, sugarbeet and wheat storage, aeration and drying solutions for improved energy efficiency (reduced GHGs emission protocol) and quality retention. Worked with farmers having 5,000 - 50,000 bushel capacity bins/open storage systems equipped with 3 - 40 hp fans, direct and indirect supplemental heater systems, mixed-flow, cross-flow and double-flow high-temperature dryers running with bio-fuels, propane and natural gas systems.
- Conducted modeling and simulation of corn, wheat, rice, and soyabean conditioning and drying process for improved process efficiency and infrastructure using LCA and TEA.
- Conducted flour quality analysis and evaluated its impact on baking properties using data-driven machine learning approach.
- Developed hyperspectral image processing protocol for real-time detection of variety, quality and adulteration in grain and pulse flour components.
- Developed automated fan control systems for grain storage bins and sugarbeet storage piles using IoT Nodes, Gateways and Cloud systems.
- Assisted in organizing two agriculture field day and gave technical talks on efficient grain handling practices. Developed extension reports for Alberta Grains and Saskatchewan Wheat Commission.

- **Mentored students, research assistants and post-docs in modeling, automation, imaging protocols, big-data and energy-efficiency analysis.**

Grant details:

- **Agriculture Funding Consertium (AFC)**: Contributed to writing a proposal *'Development of Mathematical Models to Predict Airflow Distribution and Heat Transfer in Bulk Storage of Sugar Beets and Potatoes'* (PI: Dr. Chandra B. Singh) (**CAD 401,343 Cash** and **CAD 111,000** in-kind support from Lantic Sugars Inc., AGI, Southwest Design and Construction, Alberta Sugarbeet Growers, Potato Growers of Alberta) (*Status: Funded*)

- **Canada Foundation of Innovation (CFI College grant 2023)**: Assisted in writing a **CAD 2.73 million** equipment grant with in-kind support from industries (PI: Dr. Chandra B. Singh)

- **Canola Council of Canada (CCC)**: Assisted in writing the proposal *'Energy Efficient on-Farm Canola Drying in Western Canada'* (PI: Dr. Chandra B. Singh, Co-PI: Dr. Digvir S. Jayas) (**CAD 458,264**) (*Status: Reached final evaluation stage*)

May 2016 - July 2016

Intern at Indian Institute of Technology (IIT), Kharagpur, India

Details of works:- **Developed a biodegradable film for active food packaging"**
Sponsor:- Sponsored Research and Industrial Consultancy (SRIC), IIT Kharagpur
Details of work:- Prepared a film of bio-polymer, plasticizer and an ionic solution blend. The film showed enhanced effects due to cross-linking of ionic solution and the bio-polymer. Due to the incorporation of plasticizer, the film exhibited antimicrobial effect to a certain extent and enhanced mechanical properties.

Teaching Work Experience

September 2023 - December 2023

Sessional Instructor and Subject Matter Expert (SME) at School of Agriculture (Lethbridge Polytechnic)
Details of course work:-
- Harvest and Crop Storage Technology (AGS4483-C01: 3.0 Credit): A study of current agriculture harvesting and storage equipment technology, its function and maintenance, potential and application. Real data sets are evaluated using current harvest and storage technology to make agronomic recommendations. Core engineering content also included thermodynamics, psychrometrics, heat and mass transfer and fluid dynamics
- Harvest and Crop Storage Technology Lab (AGS4483L-L01)

February 2022 - August 2022

Sessional Instructor and Subject Matter Expert (SME) at School of Agriculture (Lethbridge Polytechnic)
Details of course work:-
- Telematics and Sensors (AGS3382: 3.0 Credit): Developed and taught course module related to harvest machines and the involvement of sensors in efficient yield monitoring and decision making. Introduced a variety of sensing technologies to monitor variables affecting crop production. Focused on telematics, communication technologies and sensor integration protocols including Variable rate technologies.
- Telematics and Sensors Lab (AGS3382L)
- Harvest and Crop Storage Technology (AGS4483-C01: 3.0 Credit): A study of current agriculture harvesting and storage equipment technology, its function and maintenance, potential and application. Core engineering content also included thermodynamics, psychrometrics, heat and mass transfer and fluid dynamics.

Industry Work Experience

March 2021 - May 2021

Manufacturing Engineer at Brightlight Agribusiness, Australia

Details of work:- Developed guidelines on Almond stockpile management considering fumigation practices. Developed handheld moisture meter calibration protocols for on-field moisture tracing. Developed hull-yard temperature monitoring manual using state-of-art OPI systems. Developed almond conditioning model and quality control algorithm for Qcify in-line monitoring system. Assisted in LCA analysis of almond processing operations.

December 2015

Trainee at Alok fruit and vegetables product preservation centre, Varanasi, India

Details of work:- Developed "Amla Murraba" an Indian Gooseberry product for local farmers. Developed fact sheets suitable for adoption.

May 2015 - June 2015	Intern at Twin Engineers, Vadodara, India Worked on a project on "Infrared and Microwave Process and System Designing for Grapes and Leaves" Details of work:- Developed the drying kinetic parameters for grapes for two different drying mechanisms (Microwave and Infrared). Optimized the required temperature, power of the IR lamps and frequency of magnetron, residence time of the sample and various structural parameters for designing the combined equipment.

Current Collaborations/Completed Projects

April 2025 - Current	Remote Monitoring and Efficient Aeration for Potato Storage (Project Collaborator with Dr. SenthilKumar Thiruppathi, Industry Research Chair at University of Prince Edward Island (PEI) on a 3-year study) **Funders**: PEI Climate Challenge Fund (**CAD 30,000**), In-kind support: Cavendish Farms and OPI Systems Inc. (**CAD 11,000**) **Tasks: Co-supervising a third year undergraduate, a Masters and a PhD student on Computational Fluid Dynamics modeling of aeration during potato storage and developing control algorithm.**
October 2024 - December 2025	High-Throughput Imaging System for Early Detection of Root Rot Disease in Field Peas (Pisum sativum L.) (Image acquisition activity) **Funders**:- Alberta Pulse Growers Commission (APGC), Saskatchewan Pulse Growers (SPG), Manitoba Pulse and Soybean Growers commissions and AAFC through the Sustainable Canadian Agricultural Partnership (S-CAP)
October 2024 - December 2025	UAV-based LiDAR and Multispectral Imaging for High-throughput Phenotyping of Dry Bean Breeding Trials (Lead Researcher) **Funders**:- Alberta Pulse Growers Commission (APGC) and AAFC through the Sustainable Canadian Agricultural Partnership (S-CAP)
May 2021 - September 2024	Developing on-farm stored grain drying strategies for optimum grain quality and energy efficiency (Lead Researcher) **Funders**:- Alberta Grains, OPI Systems Inc., Results Driven Agriculture Research, Alberta Innovates, Canadian Agricultural Partnership, and Saskatchewan Wheat Development Commission
May 2021 - January 2024	Development of wireless monitoring and automated aeration control system for storing sugar beets in outdoor piles (CO-Lead Researcher) **Funders**:- Alberta Innovates, OPI Systems Inc.
May 2021 - November 2021	Improving OPI fan control systems and output database using modeling technique (Lead Researcher) **Funders**:- OPI Systems Inc., Protein Industries Canada
April 2019 - February 2021	Strategies to enhance the value of on-farm grain storage in South Australia (Lead Researcher) **Funder**:- South Australian Grain Industry Trust (SAGIT)
February 2018 - April 2021	A 3D finite volume based CFD model to improve on-farm stored grain aeration strategies under Southern Australian conditions (Lead Researcher - Ph.D.) **Funders**:- SAGIT (Recieved additional research funding of **AUD 35,000**) and UniSA **Industry Partner**:- KOTZUR PTY. LTD. (Grain Silo Manufacturing Industry)
August 2016 - April 2017	Encapsulation of water-soluble vitamin using water-in-oil-in-water type multiple emulsion-based delivery system (Lead Researcher - B.Tech.) **Funder**:- NIT Rourkela
January 2016 - May 2016	Design, Development and Testing of Chironji Shelling Machine (Lead Researcher) Details of work:- Optimised the feed rate and shelling efficiency to determine the overall machine efficiency and finalised the machine with a processing capacity of 150 kg/hr

August 2015 - February 2016

Design, Development and Testing of a Continuous Aonla Pricking Machine (Lead Researcher)

Funder:- SEED, DST New Delhi

Details of work:- Optimised the feed rate, pricking efficiency and overall physical properties of the sample for designing the processing area

Publications (Published Refereed Articles) (*corresponding author)

37 Thirugnanam, N., Thirupathi, S., **Panigrahi, S.S.**, Thiruppathi, S. (2026). Photoacoustic Imaging for Food Quality and Safety: Integrating Optical-acoustic Physics with Artificial Intelligence. Trends in Food Science and Technology, 179, 106193. **https://doi.org/10.1016/j.tifs.2026.106193**

36 Alliche, R., Alliche, S.A., Announ, M., **Panigrahi, S.S.***, Hemis, M. (2026). A fluid-structure interaction model for gamma-type free-piston stirling engines: Self-start, forced-start, and experimental validation. Applied Thermal Engineering, 306, 133147. **https://doi.org/10.1016/j.applthermaleng.2026.133147**

35 Singh, M., Gowda, N.A., **Panigrahi S.S.**, Subbiah, J. (2026) Integrated Growth and Lag Phase Data-driven Predictive Modeling of *Listeria monocytogenes* in Ready-to-Eat Meats for Quantitative Risk Assessment. International Journal of Food Microbiology. 461, 112026 **https://doi.org/10.1016/j.ijfoodmicro.2026.112026**

34 **Panigrahi, S.S.**, Thirulogasundar, A., Ramalingam, S., Singla, D., Sopiwnyk, E., Singh, C.B. (2026). Impact of On-farm High-temperature Wheat Drying Parameters on Milling Quality: Process Efficiency vs Quality Comparison. Journal of Stored Products Research. 119, 103189. **https://doi.org/10.1016/j.jspr.2026.103189**

33 **Panigrahi, S.S.**, Singh, C. B., (2026). Comparison on energy efficiency, carbon emissions, and cost implications of cross-flow, mixed-flow, and double-flow grain dryers in evaluating process sustainability. Cleaner Engineering and Technology. 34, 101289. **https://doi.org/10.1016/j.clet.2026.101289**

32 Singla, M., Olakanmi, S.J., Senthilkumar, T., **Panigrahi S.S.**, Singh C.B. (2026). Energy-Efficient Wireless Monitoring and Automated Aeration System for Sustainable Sugar Beet Pile Conditioning. Sugar Tech. **https://doi.org/10.1007/s12355-026-01773-9**

31 **Panigrahi, S.S.**, Singh, K.D., Balasubramanian, P., Wang, H., Natarajan, M. (2026). An Integrated UAV-Based Spectral-Index Fusion Framework Using Machine Learning Classifiers to Model Physiological Maturity in Dry Bean. Computers and Electronics in Agriculture. 246, 111573. **https://doi.org/10.1016/j.compag.2026.111573**

30 **Panigrahi, S.S.**, Singh, C.B. (2026). Optimization of Numerical Approach for a Commercial Grain Conditioning Simulation Tool (PHAST Model) using Finite Difference Method. Applied Engineering in Agriculture (ASABE). 42(1): 69-80. **doi: 10.13031/aea.16468**

29 Ravichandran P., Singh, K.D, Randhawa, H., **Panigrahi, S.S.** (2026). Forecasting Spring Wheat Maturity from UAV-Based Multispectral Imagery Using Machine and Deep Learning Models. AgriEngineering. 8(2), 62. **https://doi.org/10.3390/agriengineering8020062**

28 Mohammed K., Hemis, M., **Panigrahi, S.S.*** (2025). Indirect Effects of Solar Energy on Phase Change Material-Based Thermal Energy Storage Unit. International Communications in Heat and Mass Transfer. 172 (Part 2, 2026), 110333. **https://doi.org/10.1016/j.icheatmasstransfer.2025.110333**

27 Lohani, U., Senthilkumar, T., Singla, D., **Panigrahi, S.S.**, Singh, C.B. (2025). Evaluation of pasting and gelling properties of chickpea-corn flour blends. Food Hydrocolloids. 173(April 2026), 112238. **https://doi.org/10.1016/j.foodhyd.2025.112238**

26 Senthilkumar, T., **Panigrahi, S.S.**, Thirugnanam, N., Kaushik Raja, B.K.R. (2025). Automation in the Shellfish Aquaculture Sector to Ensure Sustainability and Food Security. AgriEngineering. 7(11), 387. **https://doi.org/10.3390/agriengineering7110387**

25 **Panigrahi, S.S.**, Mishra, G., Kumar, V., Panigrahi, C., Kaur, G., Sheikh, A., Kumar, S., Panda, B.K. (2025). Next-Generation Food Drying: Specialized and Smart Approaches to Boost Efficiency and Quality. Comprehensive Reviews in Food Science and Food Safety. 24(5), e70269. **https://doi.org/10.1111/1541-4337.70269**

24 **Panigrahi, S.S.**, Singh, C. B., (2025). Thermodynamic-based Theoretical Energy Efficiency Model for High-temperature Dryer: Process Efficiency Validation for Conventional and Combination Drying. Biosystems Engineering. 258 (2025), 104248. **https://doi.org/10.1016/j.biosystemseng.2025.104248**

23 Rabah, A. B., Hemis, M., Ali, L. S., **Panigrahi, S.S.***, Houria, M., Raghavan, V. (2025). Photovoltaic-thermal Hybrid Solar Drying of Chamomilla Nobile Flowers. Journal of Stored Products Research, 114 (2025), 102755. **https://doi.org/10.1016/j.jspr.2025.102755**

22 **Panigrahi, S.S.**, Hemis, M., Singh, C.B. (2025). Energy and Moisture Source-term Based Distributive (Luikov) Parameter Model to Simulate Canola Hot-air Drying. Journal of Stored Product Research, 114 (2025), 102701. **https://doi.org/10.1016/j.jspr.2025.102701**

21 **Panigrahi, S.S.**, Singh, K., Balasubramanian, P., Wang, H., Natarajan, M., Ravichandran P., (2025). UAV-based LiDAR and Multispectral Imaging for Estimating Dry Bean Plant height, Lodging and Seed yield. Sensors, 25(11), 3535. **https://doi.org/10.3390/s25113535**

20 Benyoucef, M., **Panigrahi, S.S.*** (2025). Fluidization-bed drying and microwave radiation effects on drying rate, fatty acid, protein and germination of rapeseed. Journal of Food Composition and Analysis, 47 (2025), 107562. **https://doi.org/10.1016/j.jfca.2025.107562**

19 Tellab, N., Hemis, M., **Panigrahi, S.S.**, Singh, C.B. (2025). Influence of Rewetting Method on Soybean Behaviour during Fluidized Bed Drying, Experimental and Simulation Study. Journal of Stored Product Research, 111, 102548. **https://doi.org/10.1016/j.jspr.2025.102548**

18 **Panigrahi, S.S.**, Gregor, L., Singh, C.B. (2024). Analysis of In-bin Low-temperature and High-temperature Grain Drying in Alberta: Specific Energy, Carbon Costs and Policy Implications. Energy Reports. Vol 12(2024), 3341-3352. **https://doi.org/10.1016/j.egyr.2024.09.013**

17 Omar L.B., Benyoucef, M., **Panigrahi, S.S.**, Hemis, M., Singh, C.B. (2023). Fluidization-Bed Drying and Microwave Radiation Effects on Drying rate, Fatty acid, Protein and Germination of Flaxseed. Journal of Food Process Engineering, 47(1), e14524. **https://doi.org/10.1111/jfpe.14524**

16 Poovai, P.D., Athmaselvi, K.A., Arumuganathan, T., **Panigrahi, S.S.**, Indu Rani, C. Neelavathi, R. (2023). Effects of Tray and Freeze Drying on Physico-chemical and Structural Properties of Fig Fruit Powder. Journal of Agricultural Engineering (India). Vol 60(3) July-September, 2023 **https://doi.org/10.52151/jae2023603.1811**

15 **Panigrahi, S.S.**, Luthra, K., Singh, C.B., Atungulu G., Corscadden, K., (2023). On-farm Grain Drying System Sustainability: Current Energy and Carbon Footprint Assessment with Potential Reform Measures. Sustainable Energy Technologies and Assessments. **https://doi.org/10.1016/j.seta.2023.103430**

14 Zare, D., Nourmohamadi-Moghadami, A., **Panigrahi, S.S.**, Karim, A., Singh, C.B., (2023). Dimensionless Modelling of Fine Material Distribution in an Experimental Silo during Central Spout Loading. Journal of Stored Product Research. 100, 102063. **https://doi.org/10.1016/j.jspr.2022.102063**

13 Gaurav, A., **Panigrahi, S.S.**, Pradhan, R. C., Mishra, S. (2022). Co-rotating extrusion cooking impact on product characteristics using hulled kodo millet and hybrid maize flour. Journal of Food Science and Technology, 1-11. **https://doi.org/10.1007/s13197-022-05467-4**

12 **Panigrahi, S.S.**, Singh, C. B., Fielke J. (2022). A 3D transient CFD model to predict heat and moisture transfer in on-farm stored grain silo through parallel computing using compiler directives: Impact of discretization methods on solution efficacy. Drying Technology. **https://doi.org/10.1080/07373937.2022.2121284**

11 **Panigrahi S.S.***, Fielke J.M., Singh C.B. (2022). Evaluating isotherms and isosteric heat utilization during sorption characteristics of feed barley. Journal of Stored Product Research, 96, 101955. **https://doi.org/10.1016/j.jspr.2022.101955**

10 Panda B.K., **Panigrahi S.S.**, Mishra G., Shrivastava S.L. (2021).Microwave-Assisted Hydration of Freshly Harvested Paddy (Oryza sativa L.): Process Development Based on Soaking Characterization and Energy Utilization. Food and Bioprocess Technology. **https://doi.org/10.1007/s11947-021-02682-3**

9 Panda B.K., Mishra G., **Panigrahi S.S.**, Shrivastava S.L. (2021). Microwave-assisted parboiling of high moisture paddy: A comparative study based on energy utilization, process economy and grain quality with conventional parboiling. Energy. **https://doi.org/10.1016/j.energy.2021.121011**

8 **Panigrahi S.S.**, Singh, C. B., Fielke J. (2021). Strategies to mitigate dead-zones in on-farm stored grain silos fitted with aeration ducting modelled using computational fluid dynamics. Biosystems Engineering, 205, 93-104. **https://doi.org/10.1016/j.biosystemseng.2021.02.013**

7 **Panigrahi S.S.**, Singh, C. B., Fielke J. (2020). CFD modelling of physical velocity and anisotropic resistance components in a peaked stored grain with aeration ducting systems. Computers and Electronics in Agriculture. 179, 105820. **https://doi.org/10.1016/j.compag.2020.105820**

6 Chandrasekaran, I., **Panigrahi S.S.**, Ravikanth, L., Singh, C. B. (2019). Potential of Near-Infrared (NIR) Spectroscopy and Hyperspectral Imaging for Quality and Safety Assessment of Fruits: an Overview. Food Analytical Methods, 12, 2438–2458. **https://doi.org/10.1007/s12161-019-01609-1**

5 **Panigrahi S.S.**, Singh C.B., Fielke J.M., Zare D.(2019) Modelling of heat and mass transfer within the grain storage ecosystem using numerical methods: A review. Drying Technology, 38(13), 1677-1697. **https://doi.org/10.1080/07373937.2019.1656643**

4 **Panigrahi S.S.**, Syed I., Sivabalan S. and Sarkar P. (2018). Nanoencapsulation strategies for lipid-soluble vitamins. Chemical Papers, 73, 1-16. **https://doi.org/10.1007/s11696-018-0559-7**

3 Sarkar P, Choudhary R, **Panigrahi S**, Syed I, Sivapratha S, Chanda CV (2017). Nano-inspired systems in food technology and packaging. Environmental Chemistry Letters, 15, 607-622. **https://doi.org/10.1007/s10311-017-0649-8**

2 Behera G, **Panigrahi SS**, Rayaguru K. (2016). Effect of Storage Conditions on Quality of Ready-to-Serve Beverage from Residual Osmotic Solution of Star fruit. Advances in Food Science and Engineering. **DOI: 10.22606/afse.2017.14002**

1 Sarkar P, Kumar DHL, Dhumal C, **Panigrahi SS**, Choudhary R. (2015). Traditional and ayurvedic foods of Indian origin. Journal of Ethnic Foods, 2(3), 97-109. **https://doi.org/10.1016/j.jef.2015.08.003**

Publications (Unpublished Refereed Articles) (*corresponding author)

2 **Panigrahi, S.S.**, Rahul, Singh, M., Gowda, N., Subbiah, J., (2026). From Gamma Models to Explainable AI: A Comparative Framework for Predicting Listeria monocytogenes Growth in Ready-to-Eat Meat Products. Food Research International. **Under Review**

1 Lamia, S.A., **Panigrahi, S.S.***, Rabah, A.B., Horri H., Zobir, F., Chikhi, W., Hemis, M. (2026). Effects of Infrared-assisted Hot-air Drying on the Chamomile Bioactive Substances. Food Science and BioTechnology. **Under Review**

Publications (Conference Papers)

7 Senthilkumar T, **Panigrahi S.S.**, Singla M., Singh CB (2025) Development of wireless monitoring and automated aeration control system for sugar beet conditioning in outdoor piles. In 2025 ASABE Annual International Meeting (p. 1). American Society of Agricultural and Biological Engineers. doi:10.13031/aim.202500999

6 Senthilkumar T, **Panigrahi S.S.**, Singh CB (2024) Continuous Carbon dioxide Monitoring in a Commercial Bin Stored with Canola. Pg. 220-226. In: Jayas DS, Singh CB, Jian F (eds) Proceedings of the 12th International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF 2024), CAF Permanent Committee Secretariat, Winnipeg, Canada

5 **Panigrahi S.S.**, Senthilkumar T, Singh CB (2024) Automated aeration of canola in southern Alberta (Canada). Pg. 54-60. In: Jayas DS, Singh CB, Jian F (eds) Proceedings of the 12th International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF 2024), CAF Permanent Committee Secretariat, Winnipeg, Canada

4 **Panigrahi S.S.**, Singh C.B., Fielke J.M. (2022) Evaluation of aeration strategy for on-farm stored grain in southern Australian conditions using CFD-based control system. (Paper ID: 149). CSBE/SCGAB AGM and Technical Conference 2022, Charlottetown

3 **Panigrahi S.S.**, Singh C.B., Fielke J.M. (2021). Integrated CFD-based on-farm stored grain aeration model to predict the fan hours and investigate factors affecting the cooling potential. Pp. 54-61. In: Jayas DS, Jian F (eds) Proceedings of the 11th International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF2020), CAF Permanent Committee Secretariat, Winnipeg, Canada.

2 Zare, D.,Nourmohamadi Moghadami, A., Singh, C. B., Stroshine, R. L., **Panigrahi S.S.** (2019). Modeling of fine materials distribution in an experimental silo using Dimensional Analysis. In 2019 ASABE Annual International Meeting (p. 1). American Society of Agricultural and Biological Engineers.

1 **Panigrahi S.S.**, Singh C.B., Fielke J.M. (2019) Effect of Mediterranean climatic condition during aeration and silo wall coating in on-farm grain storage in South Australia. (p. 1). American Society of Agricultural and Biological Engineers.

Publications (Published Book Chapters)

3 Panda B.K., **Panigrahi S.S.**, Mishra G., Kumar V. (2023). Robotics for general material handling machines in food plants. In. Transporting Operations of Food Materials Within Food Factories. **https://doi.org/10.1016/B978-0-12-818585-8.00005-2**

2 Garg, S., Hemrom, A., Syed, I., Sivapratha, S., **Panigrahi S.S.**, Dhumal, C., and Sarkar, P. (2019). Nanoedible Coatings for Dairy Food Matrices. Nanotechnology Applications in Dairy Science: Packaging, Processing, and Preservation, 27.

1 Sarkar P, **Panigrahi SS**, Banerjee P. (2016). Nanosensors in food safety- Book chapter. Portable Biosensors and Point-of-Care systems. **DOI: 10.1049/PBHE003E. ch10**

Publications (Unpublished Book Chapters)

2 **Panigrahi, S.S.** and Singh, C.B.,(2026). Advances in Computer Simulation and Modeling for In-bin Grain Management. In Advances in In-Bin Grain Drying System and Infrastructure. CRC Press. **Under Review**

1 Nwaizu, C., and **Panigrahi, S.S.**, Singh, C.B. (2026). Low Temperature Drying for an Advanced On-Farm In-Bin Grain Management System. In Advances in In-Bin Grain Drying System and Infrastructure. CRC Press. **In Preparation**

Technical skills

Software/ Packages	Matlab, SolidWorks, ANSYS, EDEM, OpenFoam, PyTorch, Spyder, PyCharm (Open source ML and DL modeling), OpenLCA, SuperPro Designer, High-Performance Computing, NVIDIA's PhysicsNeMo, Microsoft Azure: Digital Twins
Programming Languages	C, C++, Python, Visual Basic, R
Statistical Packages	OriginPro, Design expert, Aspen Unscrambler, SPSS, MiniTab
Laboratory Instruments	LECO Protein Analyser, Rapid Visco Analyser, Fourier Transform Infrared Spectroscopy, High-performance Liquid Chromatography, Texture analyser, Glutomatic, Falling Number, NIR grain analyser, Multispectral Camera and Hyperspectral Imaging System, Farinograph, Extensograph, Alveograph, Differential Scanning Calorimetry, Twin-screw Extruder, HTST Pasteurization system, Spray Dryer, Cold Plasma, Ozone System
Industrial Operations	**Almond Industry**: Moisture Content Programmable Computer (MCPC), Tomra Sorters, Qcify, Hulling machines, Graders and Colour Sorters; **Grain Industry**: High-temperature dryers, Low-temperature heaters, Automated fan programmable control systems, Grain monitoring sensor systems (OPI Systems, AGI SureTrack, GSI, Bin-Sense, TopGradeAg, Adaptive Agriculture)

Scholastic Achievements

August 2020 (Australia)	Voted **2020 People's Choice Award in STEM** for the 3-minutes PhD thesis presentation
March 2019 (Australia)	Received Travel funding (AUD 5,000) from **SOUTH AUSTRALIAN GRAIN INDUSTRY TRUST (SAGIT)** for the PhD
March 2019 (Australia)	Received additional research funding (AUD 35,000) from **SOUTH AUSTRALIAN GRAIN INDUSTRY TRUST (SAGIT)** for the PhD

February 2018 (Australia)	Recipient of Prestigious **AUSTRALIAN GOVERNMENT (COMMONWEALTH) RESEARCH TRAINING PROGRAM INTERNATIONAL SCHOLARSHIP** and **HIGHER DEGREE RESEARCH FEE OFFSET**
October 2014 (India)	Secured 1st Position in the Quiz competition held on WORLD FOOD DAY
May 2013 (India)	Awardee of the Prestigious **PRIME MINISTER SCHOLARSHIP** for my entire undergraduate studies

Professional/Leadership Association

Editorial Advisory Board Member (2026-	Journal of Stored Products Research
Arkansas Section of ASABE (2026-27)	Secretary
ASABE 2026-27	Respresenting PRS Community at ASABE's Strategic Initiative Council
ASABE CPD 1 Organizer 2026 AIM	Future of Stored Grain Monitoring and Management (Secured **USD 1,500** in sponsorship funding from the Grain Elevator and Processing Society (GEAPS) and Agri Process Innovations)
Topic Editor 2025-26	Frontiers in Food Science and Technology: Waste-to-Value Strategies for Developing Next-Generation Functional Food Products
Guest Editor 2025-26	Pre and Post-Harvest Engineering in Agriculture: Cutting-Edge Technologies in Grain Drying: Innovations for Efficiency, Quality, and Sustainability
Program Chair 2025-26	Processing Systems 702 Committee - Crop and Feed Processing and Storage, ASABE
President 2025-26	Association of Agricultural, Biological, and Food Engineers of Indian Origin (AABFEIO), ASABE
Chair 2025-26	Processing Systems Community, ASABE (managed 7 committees and representative groups to successfully conduct community business)
CSRS 2025	Conference Organizing Committee Member for Canadian Society of Remote Sensing (CSRS) 2025
Professional Engineer 2025-current	Association of Professional Engineers and Geoscientists of Alberta (APEGA)
ASABE Standards 2025-current	Part of a committee for updating ASABE D245.7 Standards for moisture relationship
Deputy Chair 2024	DRI EDIA Champions Pilot Program for Applied Sciences Subcommittee (Reviewed 10 proposals and participated to finalize 28 out of 43 proposals for funding)
Past-President 2024-25	Association of Agricultural, Biological, and Food Engineers of Indian Origin (AABFEIO), ASABE
Vice-Chair 2024-25	Processing Systems Community, ASABE
Engineer-in-Training 2024-25	Association of Professional Engineers and Geoscientists of Alberta (APEGA)
Guest Editor 2024-25	Advances in Sustainable Food Process Engineering, Foods
Associate Editor 2025-current	Processing Systems, ASABE Journals
Secretary 2023-24	Processing Systems Community, ASABE

President 2023-24	Association of Agricultural, Biological, and Food Engineers of Indian Origin (AABFEIO), ASABE
CSABE 2023	Conference Committee Member and Technical Program Chair
YPC ASABE 2022 - Current	Technical Committee Member
ASABE Reviewer 2022-Current	Recognized Certified Journal Reviewer
AMII 2022 - Current	Certifications from Alberta Machine Intelligence Institute (AMII) Certificate1:https://www.credential.net/1287a93b-5c43-4625-84b3-86b7fb714ac6. Certificate2:https://www.credential.net/b7fc2655-9cd5-4cfd-860f-a6a484ccb2d7.
Secretary-Elect 2022-23	Processing Systems Community, ASABE
Vice-President 2022-23	Association of Agricultural, Biological, and Food Engineers of Indian Origin (AABFEIO), ASABE
Academic Collaborator 2022	Prairie Agricultural Machinery Institute (PAMI), Canada.
Guest Editor 2021-23	American Journal of Bioscience and Bioengineering (BIO)
Editorial Board Member2021-23	Appointment as Editorial Board Member of BIO journal
ON Prime 2018-21	ON Prime5 Research Team – part of the ON accelerator, powered by CSIRO Company: Commonwealth Scientific and Industrial Research Organisation (CSIRO)
ASABE 2018-Current	American Society of Agricultural and Biological Engineers
AABFEIO 2022-Current	Association of Agricultural, Biological, and Food Engineers of Indian Origin
CSABE/SCGAB 2018-Current	Canadian Society for Agricultural and Biosystems Engineering

Conference/Meetings

November 2026	Invited Speaker at Grain Bin School, NERREC, Harrisburg, Arkansas- **Gave a talk on 'Is Your Grain Bin Talking?'**
October 2026	Arkansas Section of ASABE Annual Meeting, Fayetteville, Arkansas USA- **Master Student Presented 1 research poster on 'Ozone Treatment of Enhancing Seed and Grain Quality During Post-harvest Storage'**
October 2026	Arkansas Section of ASABE Annual Meeting, Fayetteville, Arkansas USA- **Master Student Presented 1 research poster on 'Experimental and CFD Modeling of Ozone Transport and Distribution in a Deep Grain Bed'**
July 2026	AMERICAN SOCIETY OF AGRICULTURAL AND BIOLOGICAL ENGINEERS (ASABE) Annual International Meeting, Indianapolis, USA- **Presented 1 research article on Predictive Microbiology**
August 2025	Northeast Agricultural / Biological Engineering Conference (NABEC), Cornell University, USA- **Co-authored 1 poster on "Evaluation of Pasting and Gelling Properties of Chickpea Flour Adulterated with Corn Flour"**
June 2025	CANADIAN SOCIETY OF REMOTE SENSING (CSRS), Lethbridge, Canada- **Presented 1 poster on "UAV-Based Multispectral Imaging for Estimating Dry Bean Crop Maturity"**
July 2024	CANADIAN SOCIETY OF BIOENGINEERING (CSBE), Winnipeg, Canada- **Presented 1 research article**
June 2024	Grain School Meeting Conducted by MJ Enterprises in Rosalind, Alberta, Canada- **Presented Alberta Grain drying results to industry and farmer community**

July 2023	CANADIAN SOCIETY OF BIOENGINEERING (CSBE), Lethbridge, Canada- **Presented 1 research article**
July 2023	AMERICAN SOCIETY OF AGRICULTURAL AND BIOLOGICAL ENGINEERS (ASABE) Annual International Meeting, Nebraska, USA- **Presented 1 research article**
July 2022	CANADIAN SOCIETY OF BIOENGINEERING (CSBE), Charlottetown, Canada- **Presented 1 research article**
July 2022	AMERICAN SOCIETY OF AGRICULTURAL AND BIOLOGICAL ENGINEERS (ASABE) Annual International Meeting, Houston, USA- **Presented 1 research article**
August 2021	International Conference on Controlled Atmosphere and Fumigation in Stored Products (CAF2020), CAF Permanent Committee Secretariat, Winnipeg, Canada.- **Presented 1 research article**
September 2020	SOUTH AUSTRALIAN GRAIN INDUSTRY TRUST (SAGIT) ANNUAL UPDATE, ADELAIDE SHOWGROUNDS. Interacted over future management of stored grain facilities
July 2019	LEAP AUSTRALIA PTY. LTD. Meeting and training on ANSYS Fluids at the University of Adelaide
July 2019	AMERICAN SOCIETY OF AGRICULTURAL AND BIOLOGICAL ENGINEERS (ASABE) Annual International Meeting.
July 2018	SOUTH AUSTRALIAN GRAIN INDUSTRY TRUST (SAGIT) ANNUAL UPDATE, ADELAIDE SHOWGROUNDS. Interacted with the Scientific Officer for future management of stored grain facilities
July 2018	LEAP AUSTRALIA PTY. LTD. Meeting and training on finite element analysis (FEA) and computational fluid dynamics (CFD) at the University of Adelaide
June 2018	AUSTRALIAN GRAIN STORAGE AND PROTECTION CONFERENCE, MELBOURNE. Organizer: The National Working Party on Grain Protection (GRAIN TRADE AUSTRALIA). I presented an update on **ON-FARM AERATION TRIALS** conducted in stored grain silo at Balaklava (SA)

Journal Reviewer

Journal of Stored Product Research (*Active reviewer with 40+ reviews*)
Computers and Electronics in Agriculture (*Active reviewer with 10+ reviews*)
Powder Technology
Drying Technology (*Active reviewer with 10+ reviews*)
Journal of Food Process Engineering
Measurements: Foods
Applied Food Research
Journal of the ASABE (*Active reviewer with 5+ reviews*)
Applied Engineering in Agriculture (*Active reviewer with 5+ reviews*)
Food and Bioprocess Technology
Journal of Agricultural Engineering (India)